

tend to repel one another, producing undesirable beam divergence. The unwanted effect is countered to some extent as the propagating beam may knock off electrons from air molecules, which would in turn intermingle with the beam and neutralise the charged particles. The magnetic field created by the charged-particle beam current also prevents beam spreading by producing a type of conduit. For a charged-particle beam to propagate satisfactorily through the atmosphere, the particle beam needs to have certain threshold values of beam parameters including beam current, particle energy and pulse duration.

The problem is still looking for a solution as currently no particle-beam accelerator is capable of producing a beam with the required parameters. Two important experimental programmes with particle accelerator are exploring the phenomenon of particle-beam propagation through atmosphere. One of these relates to the experiments with Advanced Test Accelerator (ATA) at Lawrence Livermore National Laboratory and the other is a joint Air Force/Sandia National Laboratories programme through the use of Radial-pulse-line Accelerator (RADLAC).

Lethality

Lethality studies play an important role in understanding the precise effect a particle beam would have on interaction with different types of target materials. The subject becomes further complex as the particle beam-matter interaction would depend upon type of particles in the beam, particle energy and beam power. Parameters determining lethality and hence the efficacy of the beam weapon include beam velocity, dwell time, rapid retargeting capability, beam penetration, ancillary kill mechanisms and all-weather capability.

A beam velocity that is much higher than the target speed simplifies the task of fixing the aim point even in the case of an evasive target. As an example, a target at a distance of 50km and moving at a supersonic speed of 6 Mach would have moved only a metre or so from the time the particle beam weapon is fired till it hits the target. This is made possible by the particle beam travelling at near speed of light of 3×10^8 km/s. The beam dwell time—the time period for which the particle beam is needed to remain on the aim point on the target to inflict intended damage—is of the order of a few microseconds in the case of endoatmospheric weapons. In this case, beam power is sufficient enough to cause almost instantaneous damage. Short dwell times may be required in case of exoatmospheric beam weapons due to comparatively lesser power of these weapons.

Rapid retargeting feature of the particle-beam weapons gives them the capability of engaging multiple targets. This is made possible as the charged-particle beam can be deflected in the desired direction, within certain limits, with the help of a changing magnetic field. Unlike high-energy laser weapons, where the laser-matter interaction effects are mainly confined to the surface, particle beams have much greater penetration. The structural damage in this case is far more severe, sometimes leading even to a catastrophic damage.

Also, in the case of laser-induced damage, the material blowing off from the target surface tends to envelop or hide the target, thereby acting as a protective shield. Particle-beam weapons with their penetrating nature do not suffer from this problem. Even target hardening through shielding materials selection proves to be ineffective.

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